

TORUP, Sweden

WMO: 026180

Lat: 56.9494N

Lon: 13.0601E

Elev: 130

StdP: 99.77

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.1	-12.0	-17.0	0.9	-14.0	-14.1	1.1	-9.9	11.4	4.9	9.4	4.6	1.1	20	0.572

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	26.8	17.5	25.0	16.7	23.3	15.8	19.2	23.6	18.1	22.3	17.2	21.3	3.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	12.9	20.2	16.7	12.1	19.5	15.7	11.3	18.5	55.0	23.8	51.7	22.2	48.8	21.3	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.2	7.0	6.1	DB	-18.1	29.6	4.1	2.3	-21.1	31.3	-23.5	32.7	-25.8	34.0	-28.7	35.7	
			WB	-18.3	20.9	4.1	1.2	-21.3	21.8	-23.7	22.5	-26.0	23.2	-28.9	24.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.2	-1.4	-1.1	1.4	6.3	10.7	13.9	16.5	16.0	12.1	7.5	3.4	0.4
	DBStd	7.32	4.60	4.21	3.87	3.59	3.56	2.80	2.77	2.54	2.90	3.54	3.83	4.91
	HDD10.0	1688	354	309	268	121	33	2	0	0	14	91	197	297
	HDD18.3	4097	613	543	527	362	238	136	73	81	188	336	447	555
	CDD10.0	662	0	0	0	10	54	119	201	187	77	13	1	0
	CDD18.3	29	0	0	0	0	1	3	16	9	0	0	0	0
	CDH23.3	376	0	0	0	4	21	77	163	106	5	0	0	0
	CDH26.7	54	0	0	0	0	1	12	27	15	0	0	0	0
Wind	WSAvg	2.7	2.8	2.9	3.0	2.8	2.7	2.6	2.5	2.4	2.4	2.7	2.8	2.9
Precipitation	PrecAvg	1148	108	82	73	59	69	93	106	117	107	118	108	108
	PrecMax	1411	231	174	194	127	128	169	225	189	241	270	178	248
	PrecMin	822	7	15	27	7	11	3	14	22	28	34	57	17
	PrecStd	172	61	46	37	32	28	43	54	46	52	58	36	62
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	8.6	13.7	22.2	25.6	28.3	29.4	28.4	24.0	17.2	12.0	9.6
		MCWB	6.9	5.6	7.6	12.9	15.4	17.6	18.9	17.9	16.2	13.2	11.3	9.2
	2%	DB	6.1	6.7	10.6	18.6	23.4	25.6	26.9	26.2	21.0	14.6	10.7	8.2
		MCWB	5.4	5.4	6.3	10.8	14.2	16.1	18.1	17.4	15.1	12.7	9.8	7.6
	5%	DB	5.1	5.4	8.2	16.1	21.2	23.2	25.1	24.0	18.9	13.4	9.6	7.3
		MCWB	4.4	4.5	5.3	9.6	13.0	15.0	17.4	16.7	14.5	12.0	8.9	6.8
	10%	DB	4.2	4.3	6.6	13.5	18.6	20.8	23.0	21.9	17.1	12.4	8.6	6.4
		MCWB	3.6	3.5	4.6	8.1	12.2	14.4	16.2	16.2	13.8	11.1	7.9	5.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.0	7.1	8.7	14.0	17.6	19.1	20.8	20.3	17.9	14.5	11.5	9.3
		MCDB	7.2	7.9	11.6	20.7	23.0	25.7	25.7	24.2	21.3	15.7	11.9	9.6
	2%	WB	5.7	5.8	7.2	11.8	15.7	17.5	19.5	18.9	16.2	13.1	10.1	7.8
		MCDB	5.9	6.5	9.3	16.9	20.7	22.9	24.1	22.7	19.0	14.1	10.5	8.1
	5%	WB	4.6	4.7	6.0	10.3	14.2	16.2	18.4	17.9	15.2	12.3	9.0	6.8
		MCDB	5.0	5.3	7.6	14.6	18.8	21.2	22.8	21.4	17.6	13.2	9.6	7.2
	10%	WB	3.8	3.6	5.0	8.9	12.8	15.1	17.4	17.0	14.4	11.4	8.0	5.9
		MCDB	4.2	4.2	6.3	12.3	17.5	19.1	21.6	20.7	16.5	12.3	8.6	6.3
Mean Daily Temperature Range	5% DB	MDBR	4.7	5.6	7.9	10.5	10.9	10.2	9.7	9.7	9.0	6.6	4.6	4.5
		MCDBR	4.0	6.1	10.6	15.6	16.0	16.2	14.0	14.1	12.2	7.3	5.0	4.2
		MCWBR	4.0	4.9	6.8	8.5	8.2	8.0	6.4	6.5	7.1	5.4	4.7	4.2
	5% WB	MCDBR	4.0	4.6	8.5	13.5	12.7	12.7	11.3	10.6	10.0	6.5	4.7	4.1
		MCWBR	4.1	4.2	6.2	7.9	7.1	7.2	6.0	5.9	6.7	5.3	4.5	4.2
Clear-Sky Solar Irradiance	taub	0.298	0.326	0.345	0.381	0.378	0.369	0.391	0.391	0.359	0.347	0.323	0.288	
	taud	2.266	2.308	2.343	2.306	2.377	2.426	2.379	2.384	2.458	2.408	2.326	2.229	
	Ebn at Noon	578	722	810	832	860	872	841	814	791	692	543	477	
	Edh at Noon	51	74	92	111	110	106	109	102	82	67	49	40	
All-Sky Solar Radiation	RadAvg	0.40	1.00	2.37	3.91	5.12	5.44	5.02	4.00	2.80	1.32	0.49	0.27	
	RadStd	0.08	0.16	0.24	0.54	0.49	0.43	0.63	0.37	0.34	0.21	0.09	0.03	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (70 neighbors)		+0.44	N/A	N/A	N/A	N/A	N/A	-126	-178	N/A	N/A

Nomenclature: See separate page